Unit Test Report

1. Introduction

The Unit Test Report documents the testing performed on the Hostel Management System. The purpose of unit testing is to verify that individual functions, modules and application components work as expected.

The tests cover student operations, authentication, fee management, room allocation, leave applications, mess management, complaints, common facilities and warden operations.

The testing was performed using the Pytest framework to identify errors and verify the expected behaviour of the application.

2. Objectives of Unit Testing

- To verify the correctness of individual application modules.
- To identify errors and unexpected behaviour.
- To validate input handling and data processing.
- To verify authentication and authorization.
- To check database operations such as insertion, retrieval and updates.
- To ensure that invalid inputs and unauthorized operations are handled correctly.

3. Testing Environment

Component	Description
Programming Language	Python
Web Framework	Flask
Testing Framework	Pytest
Database	SQLite (Testing Database)
ORM	SQLAlchemy
Operating System	macOS, Windows
Development Environment	Visual Studio Code / Terminal
Test Client	Flask Test Client

The application was tested using a separate SQLite testing database. This allowed the test cases to execute without affecting the main application database.

4. Unit Test Cases

The following test cases represent the major modules and functionalities covered by the automated test suite.

4.1 Student Authentication

Test ID	Test Scenario	Expected Result	Actual Result	Status
UT-001	Load the login page	Login page should load successfully	Login page loaded successfully	PASS
UT-002	Login using valid credentials	Student should be authenticated and redirected to the dashboard	Student was authenticated and	PASS

			redirected	
UT-003	Login using an incorrect password	System should reject the login attempt	Login attempt was rejected	PASS
UT-004	Login using an unknown email	System should reject the login attempt	Login attempt was rejected	PASS
UT-005	Logout after successful login	Student session should be terminated	Student was logged out successfully	PASS

4.2 Student Dashboard and Security

Test ID	Test Scenario	Expected Result	Actual Result	Status
UT-006	Access the dashboard after login	Dashboard should load successfully	Dashboard loaded successfully	PASS
UT-007	Verify security response headers	Required security headers should be present	Security headers were present	PASS
UT-008	Access a protected page without login	User should be redirected to the login page	User was redirected	PASS

4.3 Fee and Payment Management

Test ID	Test Scenario	Expected Result	Actual Result	Status
UT-009	View student fee details	Student should be able to view their own fees	Fee details were displayed	PASS
UT-010	Make a payment for an existing fee	Payment should be recorded successfully	Payment was recorded successfully	PASS
UT-011	Generate a payment receipt	Receipt should be displayed	Receipt was displayed	PASS
UT-012	Attempt to pay another student's fee	System should prevent unauthorized payment	Request was rejected	PASS

4.4 Leave Management

Test ID	Test Scenario	Expected Result	Actual Result	Status
UT-013	Submit a valid leave application	Leave application should be saved	Application was saved	PASS
UT-014	Submit an application with missing fields	Invalid application should be rejected	Application was rejected	PASS
UT-015	Submit an application with past dates	System should reject the application	Application was rejected	PASS
UT-016	Submit an application with an end date before the start date	System should reject the application	Application was rejected	PASS
UT-017	Cancel an eligible pending leave application	Application status should change to Cancelled	Status was updated	PASS
UT-018	Cancel another student's application	System should prevent unauthorized cancellation	Request was rejected	PASS

4.5 Common Facility Management

Test ID	Test Scenario	Expected Result	Actual Result	Status

UT-019	View available facilities	Facility details should be displayed	Facility details were displayed	PASS
UT-020	Book an available facility	Booking should be created successfully	Booking was created	PASS
UT-021	Submit a booking with missing fields	System should reject the request	Request was rejected	PASS
UT-022	Book a facility for a past date	System should reject the booking	Booking was rejected	PASS
UT-023	Book an unavailable facility	System should prevent the booking	Booking was rejected	PASS

4.6 Mess Management

Test ID	Test Scenario	Expected Result	Actual Result	Status
UT-024	View the weekly mess menu	Menu details should be displayed	Menu details were displayed	PASS
UT-025	Register for a meal	Meal registration should be recorded	Registration was recorded	PASS
UT-026	Submit invalid meal details	System should reject invalid input	Invalid input was rejected	PASS
UT-027	Access mess functionality without login	User should be redirected to login	User was redirected	PASS

4.7 Complaints and Lost and Found

Test ID	Test Scenario	Expected Result	Actual Result	Status
UT-028	Load the complaints page	Complaints page should load successfully	Page loaded successfully	PASS
UT-029	Submit a valid complaint	Complaint should be recorded	Complaint was recorded	PASS
UT-030	View lost and found items	Available item details should be displayed	Item details were displayed	PASS
UT-031	Submit a found item	Item details should be saved	Item was saved	PASS

4.8 Warden Management

Test ID	Test Scenario	Expected Result	Actual Result	Status
UT-032	Access the warden dashboard using valid credentials	Warden dashboard should load	Dashboard loaded successfully	PASS
UT-033	Allow a student to access warden-only functionality	Access should be denied	Access was denied	PASS
UT-034	Add a mess menu entry	Menu entry should be saved	Menu entry was saved	PASS
UT-035	Update an existing mess menu entry	Menu details should be updated	Menu was updated	PASS
UT-036	Add a duplicate mess menu entry	System should reject duplicate entries	Duplicate entry was rejected	PASS
UT-037	Delete a menu entry with registrations	System should prevent deletion	Deletion was prevented	PASS

Note: The above table presents representative test cases from the automated test suite. It is not an exhaustive listing of all 223 automated tests.

5. Test Execution Summary

The complete automated test suite was executed using Pytest.

Parameter	Result
Total Tests Executed	223
Tests Passed	223
Tests Failed	0
Warnings	7
Overall Result	PASS

Test Execution Result

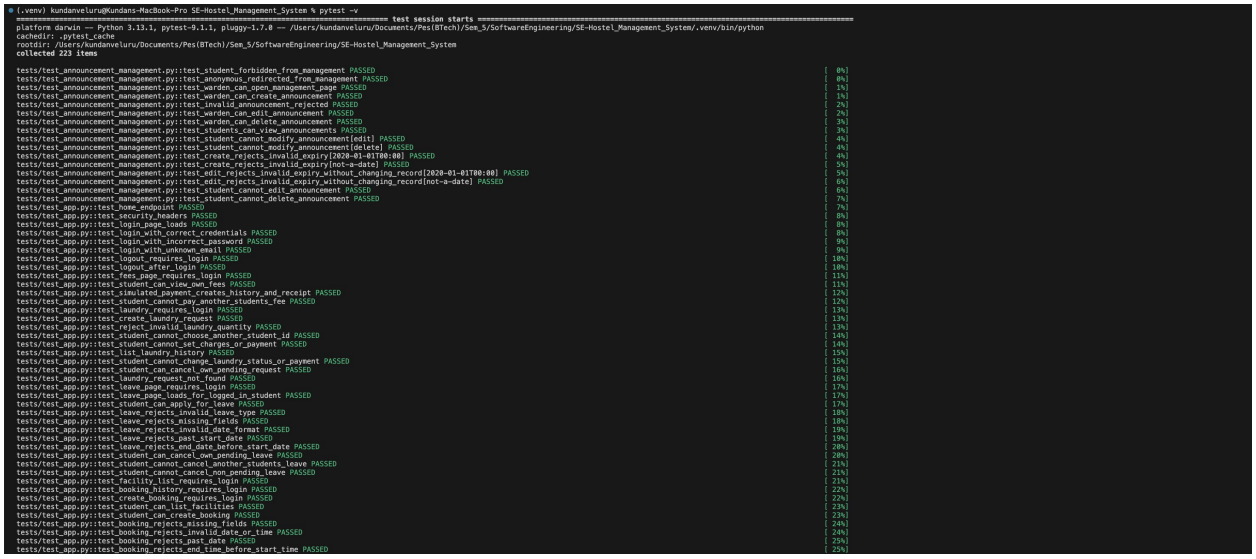
All 223 automated tests passed successfully. No test failures were reported during the final local test execution.

Seven warnings were reported during execution. These warnings did not affect the successful completion of the test suite.

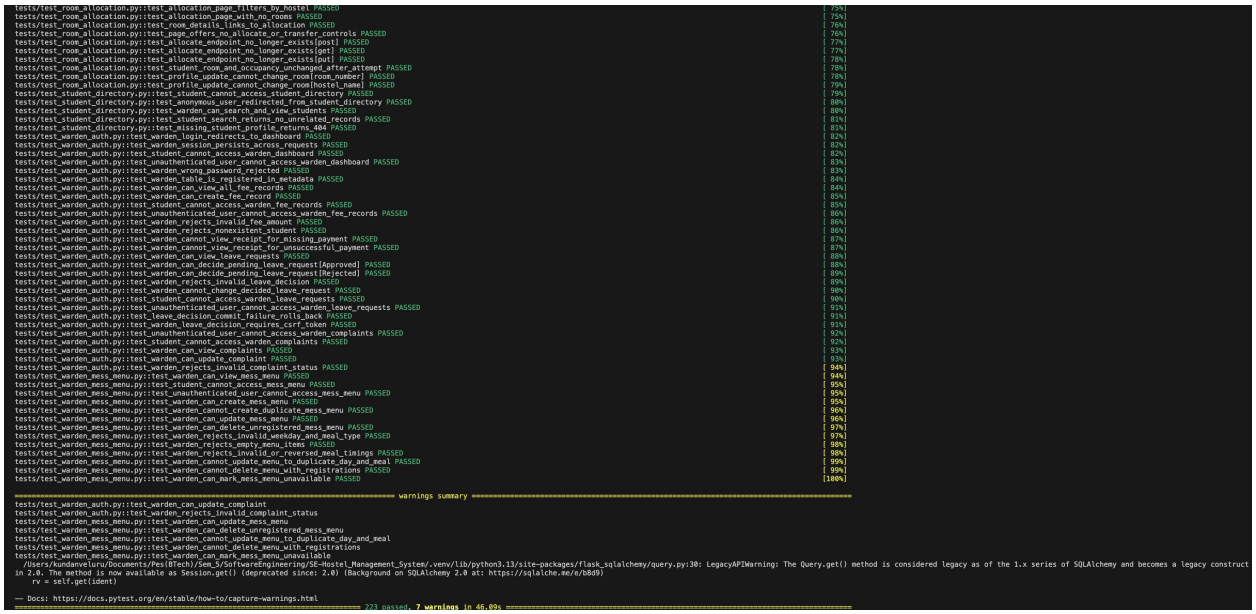
6. Testing Evidence

The following evidence can be included in the report:

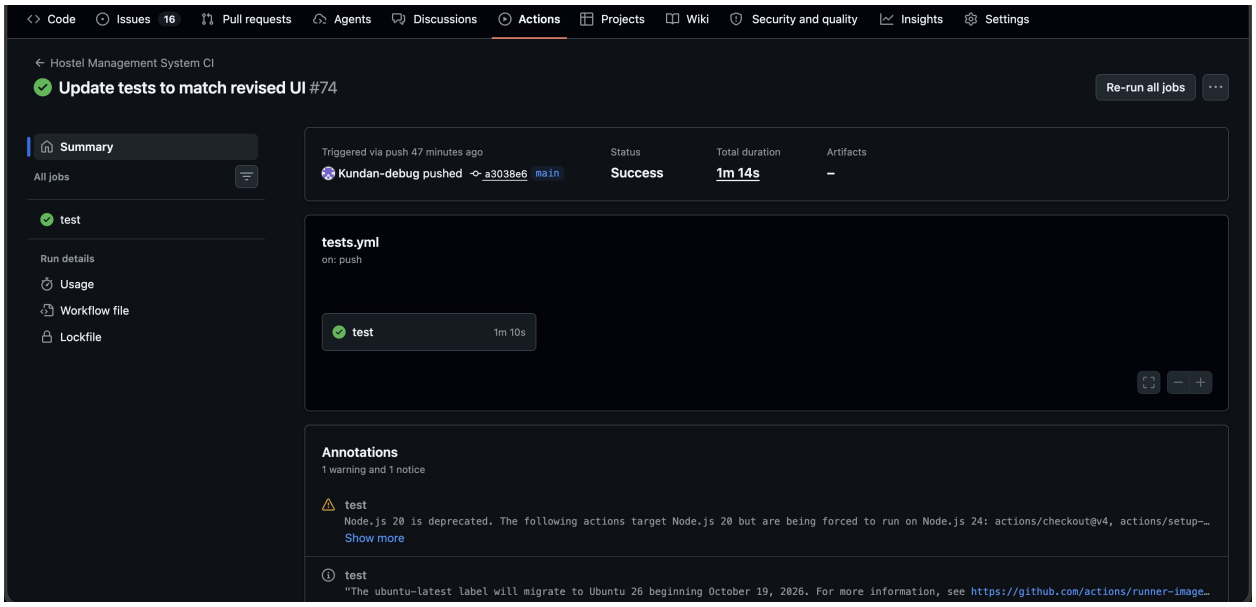
1. Screenshot of the terminal showing the execution of pytest -v.



2. Screenshot showing the final summary of 223 passed tests.



3. Screenshot of the GitHub Actions workflow showing successful execution.



Testing Command:

pytest -v

7. Conclusion

Unit testing was successfully performed on the Hostel Management System using Pytest.

The testing covered major application modules, including student authentication, dashboard access, fee and payment management, leave applications, common facilities, mess management, complaints, lost and found, and warden operations.

The final test execution resulted in 223 passed tests, zero failed tests and seven warnings.

The successful execution of the test suite indicates that the tested application functionalities are behaving as expected under the tested conditions.